

Transforms of derivatives

1. Find $L\{t \cos at\}$, assuming $L\{\sin at\} = \frac{a}{s^2 + a^2}$

ans: $\frac{s^2 - a^2}{(s^2 + a^2)^2}$

2. If $L\{\sin at\} = \frac{a}{s^2 + a^2}$, find $L\{\sin^2 at\}$

ans: $\frac{2a^2}{s(s^2 + 4a^2)}$